



Scaled Curve

A curve between two rails whose position across them changes along its length -- 30 % at the start, 70 % at the end, along a chosen transition.



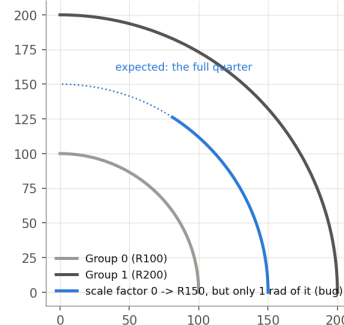
What it does

A mid-curve between two rails at a varying percentage is tedious by hand. Scaled Curve blends the rails point by point with a factor that travels from the initial to the final value.

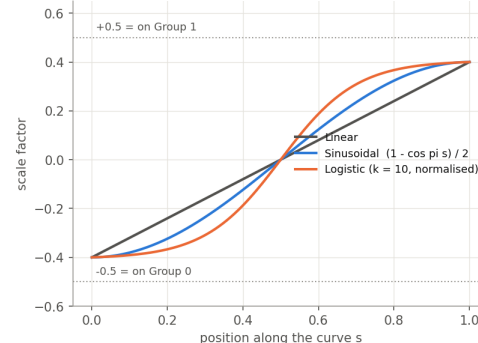
Use it when

- A character line drifting between two edges
- A midline between two rails
- Seed curves for a network

SC3: halfway between two arcs is an exact R150 arc



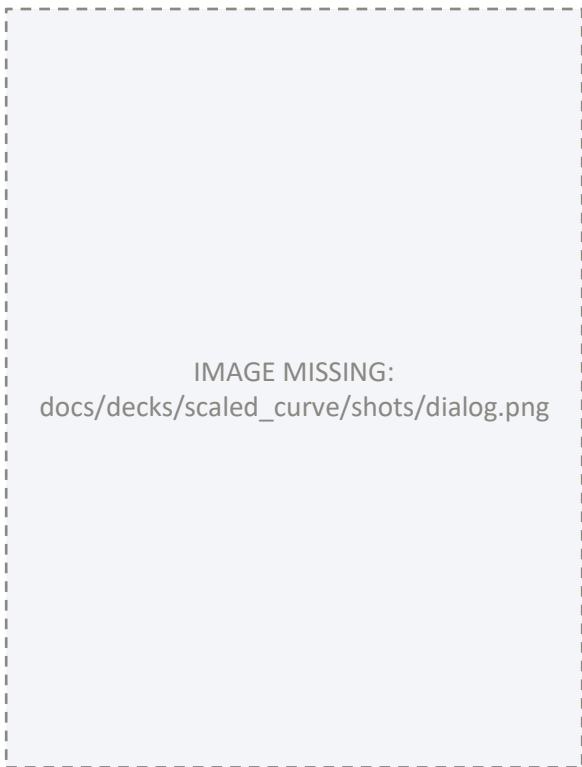
Initial -0.4 -> final +0.4: how the blend travels



SC3 and the three transition shapes.



The dialog

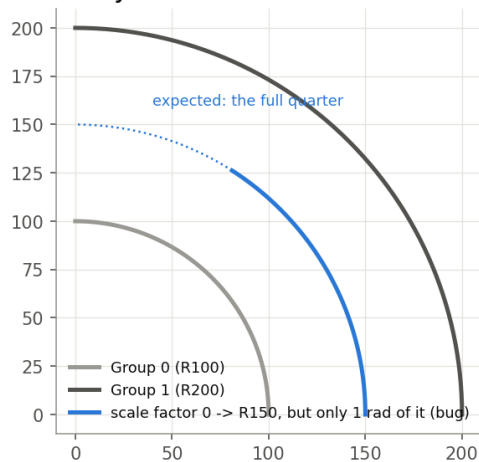


#	Parameter	What it means
1	Group 0	First rail (edges joined); Flip? reverses it.
2	Group 1	Second rail.
3	Initial curve scalefactor	-0.5 = on Group 0, 0 = halfway, +0.5 = on Group 1.
4	Final curve scalefactor	The value at the end.
5	Transition Type	Linear / Sinusoidal / Logistic.
6	Curve name	Names the wire.
7	Project onto surface?	+ face: drop the result onto it.
8	Debug & Details	Show ends / curves; output samples, tolerance, degree; input sampling.

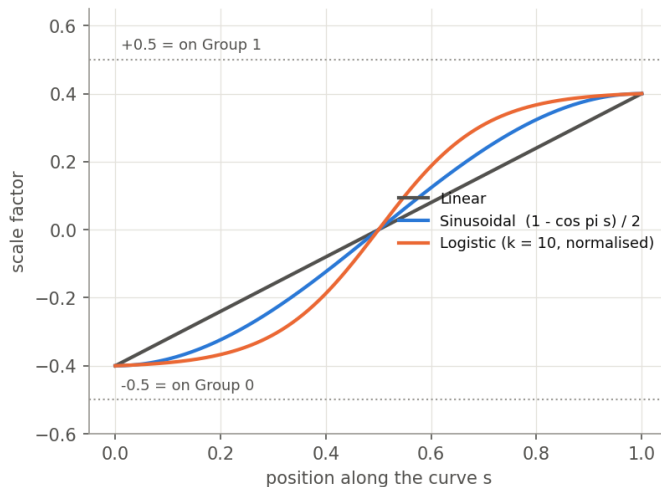


Example: SC3

SC3: halfway between two arcs is an exact R150 arc



Initial -0.4 -> final +0.4: how the blend travels



Arcs R100 and R200, scale factor 0

An exact R150 arc (within 0.00062 mm) -- but only its first radian: the known arc bug. SC0: lines $y = 0$ and $y = 100 \rightarrow y = 50$; SC2: output degree 5.



Outputs and messages

Outputs

Key	What it holds
(wire)	The scaled curve, named by Curve name.

Messages

	Message / what to do
(none)	



Tips and limits



Arcs: first radian only

On arc rails the result stops after 1 rad of each arc (bug; SC3 checks only the radius).



Matched by parameter

Not by length: uneven rails skew the blend. Orientation is manual (Flip).



Projection

Can move the ends.



Where it lives

Document	<code>gordonSurface (d0950ed72a894fbe35c27d43)</code>
Tab	<code>scaledCurve</code>
Tests	Gordon tools tests: SC0-SC3
Examples	Test Part Studio 1: Scaled Curve 1
Explainer	<code>docs/explainers/gordon_surface/gordon_surface_explained.md</code> , section 2.3
Related	Gordon Surface, Interior curves, Scaled Curve, Modify curve end, Pull surface, Simplify surface